

BIHAR BOARD – 2015

Maths

Time: 3hrs 15minutes

Max Marks: 100

1. α and β are zeros of $x^2 + x - 2 = 0$, then the value of $\frac{1}{\alpha} + \frac{1}{\beta}$ is?

- (a) $\frac{1}{2}$ (b) $-\frac{1}{\sqrt{2}}$ (c) 1 (d) 2

Ans. $\frac{1}{2}$

2. If a and b are the zeros of polynomial $p(x) = x^2 - 2x + 5$, then value of ab is-

- (a) 5 (b) -5 (c) 2 (d) -2

Ans. 5

3. How many rational numbers can occur between two rational numbers?

- (a) 1 (b) 2 (c) 3 (d) Infinite

Ans. Infinite

4. The mode of any frequency is?

Ans. Maximum frequency value.

5. If all the sides of a parallelogram touch a circle, then what is the quadrilateral?

- (a) Rectangle (b) Square (c) Rhombus (d) Trapezium

Ans. Rhombus

6. Distance of point (2, 3) from the original point would be?

- (a) 2 (b) $2\sqrt{3}$ (c) $\sqrt{13}$ (d) None of these.

Ans. $\sqrt{13}$

7. If the volume of a hemisphere is 19404 cm^3 , then the total surface area of the hemisphere is ?

- (a) 4158 cm^2 (b) 16632 cm^2 (c) 8316 cm^2 (d) 3696 cm^2

Ans. 4158cm^2

8. The probability of a certain event is-

- (a) 1 (b) 2 (c) 0 (d) None of these

Ans. 1

9. Value of $1 + \tan^2 \theta$ is –

- (a) $\sec^2 \theta$ (b) $\operatorname{cosec}^2 \theta$ (c) $\tan^2 \theta$ (d) $\cot^2 \theta$

Ans. $\sec^2 \theta$

10. $\frac{1}{\sqrt{3}}$ is the value of –

- (a) $\sin \frac{\pi}{3}$ (b) $6 \cot \frac{\pi}{3}$ (c) $\tan \frac{\pi}{6}$ (d) $\cos \frac{\pi}{6}$

Ans. $\tan \frac{\pi}{6}$

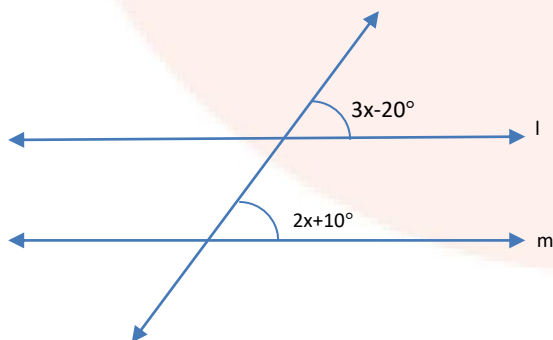
11. π is _____ number.

Ans. π is an irrational number.

12. The four degree polynomials are called _____.

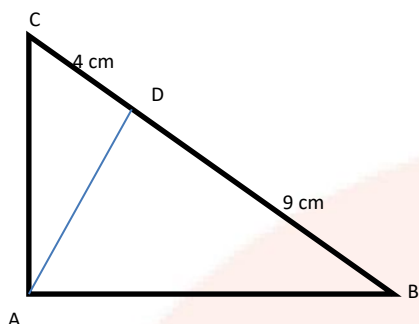
Ans. The four degree polynomials are called quadratic polynomials.

13. For _____ value of x, $l \parallel m$.



Ans. 30°

14. In the given figure, in $\triangle ABC$, $\angle A = 90^\circ$ and $AD \perp BC$. If $BD = 9\text{cm}$ and $CD = 4\text{cm}$, then $AD = \underline{\hspace{2cm}}$.



Ans. $AD = 6\text{cm}$

15. $P(-x, y)$ is $\underline{\hspace{2cm}}$ away from the origin.

Ans. The distance between $P(-x, y)$ and the origin is $\sqrt{x^2 + y^2}$.

16. What will be the value of abscissa of any point $(-4, 10)$ in a Cartesian system?

Ans. -4

17. If $2\cos A = \sqrt{3}$, then $A = \underline{\hspace{2cm}}$.

Ans. $A = 30^\circ$

18. If $3\theta = 90^\circ$, then $\tan \theta = \underline{\hspace{2cm}}$.

Ans. $\tan \theta = \sqrt{3}$

19. When the terms of any series are written under any rule, it is called

Ans. Sequence

20. The class with maximum frequency in the distribution is called class.

Ans. Modal Class

21. Find the quadratic polynomial, if the sum and product of its zeros are -3 and 2 respectively.

Ans. Let the polynomial be $ax^2 + bx + c$, and its zeroes be δ and β

According to question,

Sum of zeros = -3

$$\delta + \beta = -3 = -\frac{3}{1} = -\frac{b}{a}$$

And,

$$\delta \cdot \beta = 2 = \frac{2}{1} = \frac{c}{a}$$

If $a = 1$, then $b = 3$, $c = 2$

Therefore, the quadratic polynomial is $x^2 + 3x + 2$

22. Find the nature of the roots if the quadratic equation is $2x^2 - 6x + 3 = 0$.

Ans. Given equation is $2x^2 - 6x + 3 = 0$

Compared it with the general form $ax^2 + bx + c = 0$,

we get $a = 2$, $b = -6$ and $c = 3$.

Now, let us find the value of the discriminant ' $b^2 - 4ac$ '

$$\begin{aligned} b^2 - 4ac &= (-6)^2 - 4 \times 2 \times 3 \\ &= 36 - 24 \\ &= 12 \end{aligned}$$

$\therefore b^2 - 4ac = 12 (>0 \text{ but not a perfect square})$

Hence, the roots are real, distinct and irrational.

23. If $\triangle ABC$ is an equilateral triangle whose each side is 'a' cm, then prove that the altitude of the given triangle will be $\frac{\sqrt{3}}{2}a$.

Ans. Here, in $\triangle ADC$,

Applying Pythagoras theorem, we get

$$AD^2 + CD^2 = AC^2$$

$$\Rightarrow AD^2 = AC^2 - CD^2$$

$$\Rightarrow AD^2 = a^2 - \left(\frac{a}{2}\right)^2$$

$$\Rightarrow AD^2 = a^2 - \frac{a^2}{4}$$

$$\Rightarrow AD^2 = \frac{4a^2 - a^2}{4}$$

$$\Rightarrow AD = \sqrt{\frac{4a^2 - a^2}{4}}$$

$$\therefore AD = \frac{\sqrt{3}}{2}a$$

Hence, it is proved that the altitude of the triangle will be $\frac{\sqrt{3}}{2}a$.

24. The length of a tangent drawn from a point A at a distance of 5cm from the centre of the circle is 4cm. Find the radius of the circle.

Ans. Given,

AB is a tangent drawn on the circle from point A.

$\therefore OB \perp AB$

OA = 5cm and AB = 4 cm (Given)

In ΔABO ,

By Pythagoras theorem in ΔABO ,

$$OA^2 = OB^2 + AB^2$$

$$\Rightarrow 5^2 = OB^2 + 4^2$$

$$\Rightarrow OB^2 = 25 - 16$$

$$\Rightarrow OB^2 = 9$$

$$\therefore OB = 3$$

Therefore, the radius of the circle is 3 cm.

25. Find the distance between point $(a \cos \theta, 0)$ and $(0, a \sin \theta)$.

Ans. Let the given points be A($a \cos \theta, 0$) and B($0, a \sin \theta$).

$$\sqrt{(a \cos \theta - 0)^2 + (0 - a \sin \theta)^2}$$

$$= \sqrt{a^2 \cos^2 \theta + a^2 \sin^2 \theta}$$

$$\text{Then, } AB = \sqrt{a^2 (\cos^2 \theta + \sin^2 \theta)}$$

$$= \sqrt{a^2}$$

$$= a$$

Hence, the distance between the given points is a.

26. If the area and circumference of a circle are numerically equal, then what is the radius of the circle?

Ans. Let the radius of the circle be r units.

Then,

Perimeter/ Circumference of the circle = $2\pi r$ units
and

Area of the circle = πr^2 units

According to question,

Numerical value of area of the circle = Numerical value of perimeter of circle

So, $\pi r^2 = 2\pi r$

$\therefore r = 2$

Hence, the radius of the circle is 2units.

27. Find the volume of the right circular cone, whose height is 28cm and radius of base is 7cm.

Ans. Given,

Height of the cone (h) = 28cm

Radius of the cone (r) = 7cm

We know,

$$\begin{aligned}\text{Volume of cone} &= \frac{1}{3}\pi r^2 h \\ &= \frac{1}{3} \times \frac{22}{7} \times (7)^2 \times 28 \\ &= \frac{1}{3} \times \frac{22}{7} \times 7 \times 7 \times 28 \\ &= \frac{4312}{3} \\ &= 1437.33\text{cm}^3\end{aligned}$$

28. If point (0,2) is equidistant from the point (3, k) and (k, 5), then find the value of k.

Ans. Let the points be P(0,2) equidistant from A(3, k) and B (k,5).

According to question,

$$PA = PB$$

$$\Rightarrow PA^2 = PB^2$$

$$\Rightarrow \sqrt{(3-0)^2 + (k-2)^2} = \sqrt{(k-0)^2 + (5-2)^2} \quad [\text{Using formula for distance between two points}]$$

$$\Rightarrow 9 + (k-2)^2 = k^2 + 9$$

$$\Rightarrow (k-2)^2 = k^2$$

$$\Rightarrow k^2 - 4k + 4 = k^2$$

$$\Rightarrow -4k + 4 = 0$$

$$\Rightarrow 4 = 4k$$

$$\therefore k = 1$$

29. If $\theta = 30^\circ$, then prove that $4\cos^3\theta - 3\cos\theta = 0$.

Ans. Given, $\theta = 30^\circ$

Taking L.H.S =

$$\begin{aligned} & 4\cos^3\theta - 3\cos\theta \\ &= 4\cos^3 30^\circ - 3\cos 30^\circ \\ &= 4 \times \left(\frac{\sqrt{3}}{2}\right)^3 - 3 \times \frac{\sqrt{3}}{2} \\ &= 4 \times \frac{3\sqrt{3}}{8} - \frac{3\sqrt{3}}{2} \\ &= \frac{3\sqrt{3}}{2} - \frac{3\sqrt{3}}{2} \\ &= 0 \\ &= R.H.S. \text{proved} \end{aligned}$$

30. Find trigonometric ratios of 45° .

Ans. Let a triangle ABC be isosceles triangle where $\angle B = 90^\circ$, $\angle A = 45^\circ$ & $\angle C = 45^\circ$ and equal sides of 1 unit each.

Now, taking 45° as base angle, we get,

$$p=1$$

$$b=1$$

Using Pythagoras theorem,

$$h = \sqrt{p^2 + b^2} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\therefore \sin 45^\circ = \frac{p}{h} = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{b}{h} = \frac{1}{\sqrt{2}}$$

$$\tan 45^\circ = \frac{p}{b} = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}} = 1$$

31. Write the first term and common difference of A.P 0.6 , 1.7 , 2.8 , 3.9,....

Ans. Given A.P is 0.6 , 1.7 , 2.8 , 3.9,....

Here, the first term = 0.6

Common difference = Second term – first term

$$= 1.7 - 0.6$$

$$= 1.1$$

32. Prove that $\sqrt{5}$ is irrational number.

Ans. Lets assume that $\sqrt{5}$ is a rational number.

We know that the rational numbers should be in the form of $\frac{p}{q}$ where p and q are integers.

So,

$$\sqrt{5} = \frac{p}{q}$$

$$\Rightarrow p = \sqrt{5} \times q$$

Now,

We know that 'p' is a rational number.

So, $\sqrt{5} \times q$ should be normal as it is equal to p.

But it did not happen with $\sqrt{5}$ because it is not an integer.

Therefore, $p \neq \sqrt{5}q$

This denies that $\sqrt{5}$ is a rational number.

Hence, our assumption is wrong and $\sqrt{5}$ is an irrational number.

33. Two containers have 405 liters and 735 liters of milk. What will be the maximum capacity of the container, which can fully measure the milk of both the containers.

Ans. Given Capacity of each container is 405 and 735 litres.

Maximum capacity of required container = H.C.F (405, 735)

So, for H.C.F

$$405 = 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 5$$

$$735 = 3 \times 5 \times 7 \times 7$$

$$\therefore \text{H.C.F (405, 735)} = 3 \times 5 = 15$$

Hence, Capacity of required container is 15 litres.

34. Solve: $\sqrt{13-x^2} = x+5$

Ans.

$$\sqrt{13-x^2} = x+5$$

Squaring both the sides, we get

$$13-x^2 = (x+5)^2$$

$$\Rightarrow 13-x^2 = x^2 + 10x + 25$$

$$\Rightarrow x^2 + 10x + x^2 = -25 + 13$$

$$\Rightarrow 2x^2 + 10x = -12$$

$$\Rightarrow 2(x^2 + 5x) = -12$$

$$\Rightarrow x^2 + 5x = -6$$

$$\Rightarrow x^2 + 5x + 6 = 0$$

$$\Rightarrow x^2 + 3x + 2x - 6 = 0$$

$$\Rightarrow x(x+3) + 2(x+3) = 0$$

$$\Rightarrow (x+3)(x+2) = 0$$

Either,

$$x+3 = 0$$

$$\therefore x = -3$$

Or,

$$x+2 = 0$$

$$\therefore x = -2$$

35. For what value of k, the roots of the quadratic equation $2x^2 + kx + 3 = 0$ real and equal?

Ans. Given equation is $2x^2 + kx + 3 = 0$

We know that,

If the roots of the equation are real and equal

then, the discriminant (D) = 0

$$\Rightarrow b^2 - 4ac = 0 \dots\dots(i)$$

Comparing the given equation with $ax^2 + bx + c = 0$, we get

$$a = 2, b = k, c = 3$$

Now, putting these values in (i), we get

$$(k)^2 - 4 \times 2 \times 3 = 0$$

$$\Rightarrow k^2 - 24 = 0$$

$$\Rightarrow k^2 = 24$$

$$\Rightarrow k = \sqrt{24}$$

$$\therefore k = \pm 2\sqrt{6}$$

36. A vertical pole of a length 6 m casts a shadow 4m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.

Ans. Given,

Length of the vertical pole = 6m

Shadow of the pole = 4 m

Length of shadow of the tower = 28 m

Let Height of tower = h m

Now,

In $\triangle ABC$ and $\triangle DEF$,

$$\angle C = \angle F \quad [\text{angular elevation of sun}]$$

$$\angle B = \angle E = 90^\circ$$

$$\therefore \triangle ABC \sim \triangle DEF \text{ [By AA similarity criterion]}$$

$$\text{So, } \frac{AB}{DE} = \frac{BC}{EF} \quad [\because \text{If two triangles are similar, corresponding sides are proportional}]$$

$$\Rightarrow \frac{6}{h} = \frac{4}{28}$$

$$\Rightarrow 4 \times h = 6 \times 28$$

$$\Rightarrow h = \frac{6 \times 28}{4}$$

$$\therefore h = 42$$

Hence, the required height of the tower is 42 m.

37. If $\frac{1}{a} + \frac{1}{b} = \frac{1}{3}$, then prove that points (3,3), (a,0) and (0,b) are collinear.

38. There are 5 red and some blue balls in a bag. If the probability of drawing of a blue ball is double of the probability of drawing the red ball, then what will be the number of blue balls in the bag?

Ans. Let the number of blue balls be x .

Number of red balls = 5

Total number of balls = $x + 5$

$$P(\text{getting a red ball}) = \frac{5}{x+5}$$

$$P(\text{getting a blue ball}) = \frac{x}{x+5}$$

According to question,

$$2\left(\frac{5}{x+5}\right) = \frac{x}{x+5}$$

$$\Rightarrow 10(x+5) = x(x+5)$$

$$\Rightarrow 10x + 50 = x^2 + 5x$$

$$\Rightarrow x^2 + 5x - 10x - 50 = 0$$

$$\Rightarrow x(x+5) - 10(x+5) = 0$$

$$\Rightarrow (x+5)(x-10) = 0$$

Either,

$$x+5=0$$

$$\therefore x=-5$$

Or,

$$x-10=0$$

$$\therefore x=10$$

Since, number of balls cannot be negative,

Therefore, the number of blue balls is 10.

39. If $\pi = 180^\circ$, then evaluate $\sin^2 \frac{\pi}{3} + \tan^2 \frac{\pi}{6} - \cos^2 \frac{\pi}{2}$.

Ans. Given, $\pi = 180^\circ$

$$\begin{aligned} & \sin^2 \frac{\pi}{3} + \tan^2 \frac{\pi}{6} - \cos^2 \frac{\pi}{2} \\ &= \sin^2 \frac{180^\circ}{3} + \tan^2 \frac{180^\circ}{6} - \cos^2 \frac{180^\circ}{2} \\ &= \sin^2 60^\circ + \tan^2 30^\circ - \cos^2 90^\circ \\ &= \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{\sqrt{3}}\right)^2 - (0)^2 \\ &= \frac{3}{4} + \frac{1}{3} \\ &= \frac{9+4}{12} \\ &= \frac{13}{12} \end{aligned}$$

40. Prove that $(\cos\theta + \sin\theta)^2 + (\cos\theta - \sin\theta)^2 = 2$

Ans.

L.H.S=

$$\begin{aligned} & (\cos\theta + \sin\theta)^2 + (\cos\theta - \sin\theta)^2 \\ &= \cos^2\theta + 2\sin\theta.\cos\theta + \sin^2\theta + \cos^2\theta - 2\sin\theta.\cos\theta + \sin^2\theta \left[\begin{array}{l} \because (a+b)^2 = a^2 + 2ab + b^2 \\ (a-b)^2 = a^2 - 2ab + b^2 \end{array} \right] \\ &= \cos^2\theta + \sin^2\theta + \cos^2\theta + \sin^2\theta \\ &= 1+1 \quad [\because \sin^2\theta + \cos^2\theta = 1] \\ &= 2 \\ &= R.H.S \end{aligned}$$

41. Find median of the following distribution:

Class-interval	0-10	10-20	20-30	30-40	40-50
Frequency	10	5	8	13	14

Ans.

Class-interval	Frequency(f)	Cumulative frequency(c.f)
0-10	10	10
10-20	5	15
20-30	8	23
30-40	13	36
40-50	14	50
	N= 50	

Here, $\frac{N}{2} = \frac{50}{2} = 25$

So, the median class interval is 30-40

$\therefore l = 30$

$C = 23$

$f = 13$

$i = 10$

We know,

$$\text{Median} = l + \left(\frac{\frac{N}{2} - C}{f} \right) \times i$$

$$30 + \left(\frac{25 - 23}{13} \right) \times 10$$

$$= 30 + \frac{2}{13} \times 10$$

$$= 30 + 1.54$$

$$= 31.54$$

Therefore, median is 31.54 approximately.

42. If mean of the following distribution is 50, find the missing frequency 'f'.

Class-interval	0-20	20-40	40-60	60-80	80-100
Frequency	17	F	32	24	19

Ans.

Class-interval	Mid-point(x)	Frequency(f)	fx
0-20	10	17	170
20-40	30	F	30F
40-60	50	32	1600
60-80	70	24	1680
80-100	90	19	1710
		$\sum f = 92 + F$	$\sum fx = 5160 + 30F$

We know,

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

$$\Rightarrow 50 = \frac{5160 + 30F}{92 + F}$$

$$\Rightarrow 50(92 + F) = 5160 + 30F$$

$$\Rightarrow 4600 + 50F = 5160 + 30F$$

$$\Rightarrow 50F - 30F = 5160 - 4600$$

$$\Rightarrow 20F = 560$$

$$\Rightarrow F = \frac{560}{20}$$

$$\therefore F = 28$$

Hence, the missing frequency is 28.

43. A drinking glass is in the shape of frustum of a cone of height 45cm. If the radii of its two circular ends are 28cm and 7cm respectively, find the capacity of the glass.

Ans. Given,

Radius of two circular ends are $R = 28\text{cm}$ and $r = 7\text{cm}$

Height (h) = 45cm

Now,

Capacity of glass = Volume of frustum

$$\begin{aligned} & \frac{1}{3} \pi [R^2 + r^2 + R.r] h \\ &= \frac{1}{3} \times \frac{22}{7} \times 45 [(28)^2 + (7)^2 + 28 \times 7] \\ &= \frac{22}{7} \times 15 [784 + 49 + 196] \\ &= \frac{22}{7} \times 15 \times 1029 \\ &= 48510 \text{ cm}^3 \end{aligned}$$

Hence, the capacity of glass is 48510 cm^3

44. Solve graphically:

$$5x - y - 5 = 0 \text{ and}$$

$$3x - y - 3 = 0$$

Ans:

$$5x - y - 5 = 0$$

$$y = 5x - 5$$

Now, we will find the values of y for different values of x

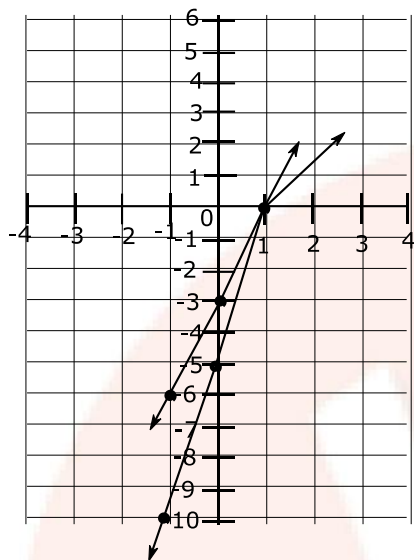
x	-1	0	1
y	-10	-5	0

$$3x - y - 3 = 0$$

$$y = 3x - 3$$

Now, we will find the values of y for different values of x

x	-1	0	1
y	-6	-3	0



Solution Point = (1,0)

45. The length of a rope between a kite and a point on the earth is 90 m. If an angle θ is formed from the ground such that $\tan\theta = \frac{15}{8}$, then find the height of the kite.

Ans. Here, $\tan\theta = \frac{15}{8}$,

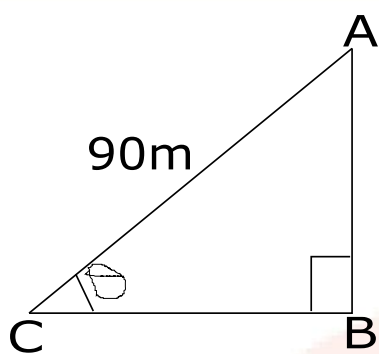
$$\Rightarrow \frac{\text{Perpendicular}}{\text{Base}} = \frac{15}{8}$$

Now,

Let perpendicular (AB) = (15x)m

Let base (BC) = (8x)m

In right angle $\triangle ABC$,



Using Pythagoras theorem,

$$AB^2 + BC^2 = AC^2$$

$$\Rightarrow (15x)^2 + (8x)^2 = (90)^2$$

$$\Rightarrow 225x^2 + 64x^2 = 8100$$

$$\Rightarrow 289x^2 = 8100$$

$$\Rightarrow x^2 = \frac{8100}{289}$$

$$\Rightarrow x = \sqrt{\frac{8100}{289}}$$

$$\therefore x = \frac{90}{17}$$

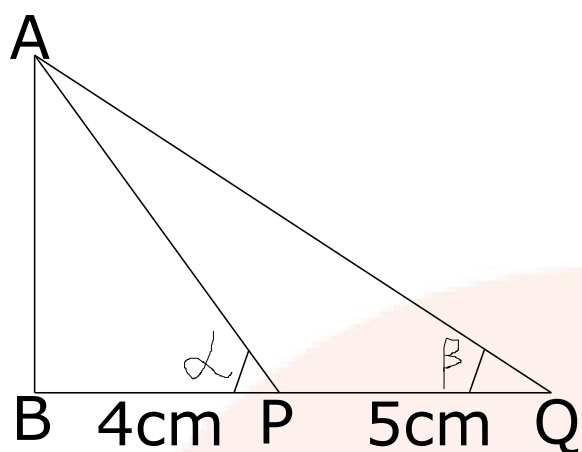
Therefore, height of kite from the ground = $(15x)m = 15 \times \frac{90}{17} = 79.4 \text{ cm}$

OR

The angle of elevation of the top of a tower from two points at a distance of 4m and 9m from the base of the tower and in the same straight line with it are complementary. Find the height of the tower.

Ans. Let AB be the tower.

P and Q be the points at distance of 4m and 9m respectively.



From fig, $PB = 4\text{m}$, $BQ = 9\text{m}$.

Let angle of elevation from P be α and angle of elevation from Q be β .

According to question,

α and β are supplementary.

$$\therefore \alpha + \beta = 90^\circ$$

In triangle ABP,

$$\tan \alpha = \frac{AB}{BP} \dots\dots\dots(i)$$

In triangle ABQ,

$$\tan \beta = \frac{AB}{BQ}$$

$$\Rightarrow \tan(90^\circ - \alpha) = \frac{AB}{BQ} \quad (\because \alpha + \beta = 90)$$

$$\Rightarrow \cot \alpha = \frac{AB}{BQ}$$

$$\Rightarrow \frac{1}{\tan \alpha} = \frac{AB}{BQ}$$

$$\Rightarrow \tan \alpha = \frac{BQ}{AB} \dots\dots\dots(ii)$$

Now,

From (i) and (ii), we get

$$\frac{AB}{PB} = \frac{BQ}{AB}$$

$$\Rightarrow AB^2 = BQ \times PB$$

$$\Rightarrow AB^2 = 4 \times 9$$

$$\Rightarrow AB^2 = 36$$

$$\therefore AB = 6$$

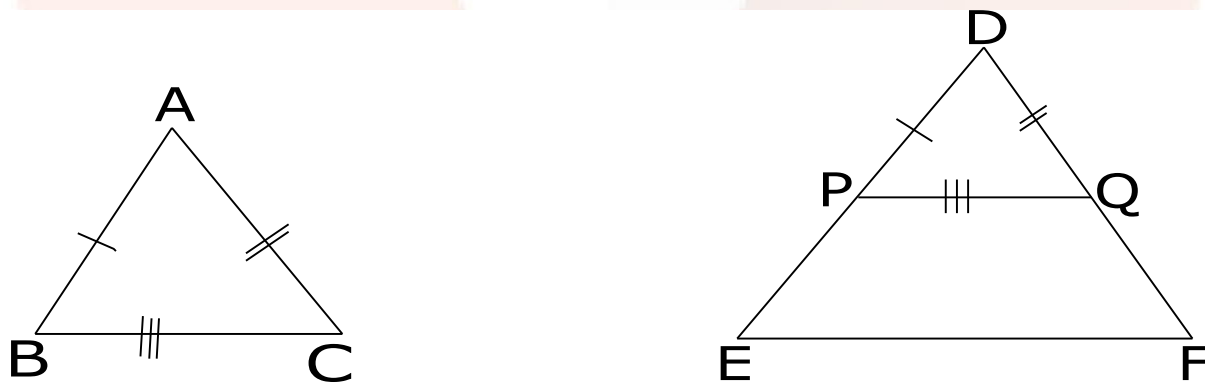
Hence, height of tower is 6m.

46. Prove that two triangles are similar if the angle of one of the two triangles is equal to the angle of the other and the corresponding sides are proportional.

Ans. Lets take two triangles ABC and DEF in which one angle is equal and their corresponding sides are proportionate.

To prove : $\triangle ABC \sim \triangle DEF$

Construction : Draw PQ in triangle DEF so that, $AB = DP$ and $AC = DF$



Here,

$\triangle ABC \cong \triangle DPQ$ [$\because$ Corresponding sides of these two triangles are equal]

$$\frac{AB}{DE} = \frac{AC}{DF} \quad [\text{Given}]$$

$$\angle A = \angle D \quad [\text{Given}]$$

Hence,

$$\frac{AB}{DE} = \frac{BC}{EF} \quad [\text{from SSS criiterion}]$$

$$\therefore \frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$$

Hence, it is proved that $\triangle ABC \sim \triangle DEF$.

OR

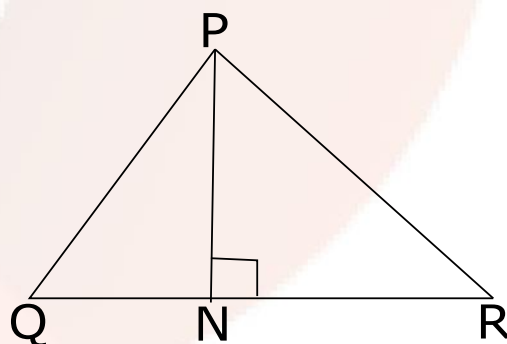
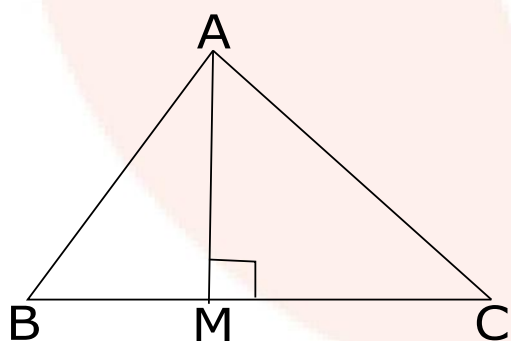
Prove that the ratio of the areas of two similar triangles is equal to the ratio of the squares of their corresponding sides.

Ans. Lets take two triangles ABC and PQR which are similar triangles.

Given, $\triangle ABC \sim \triangle PQR$

$$\text{To prove: } \frac{\text{ar}(ABC)}{\text{ar}(PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{BC}{QR}\right)^2 = \left(\frac{AC}{PR}\right)^2$$

Construction: Draw $AM \perp BC$ and $PN \perp QR$



Now,

$$ar(ABC) = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times BC \times AM \dots\dots\dots(i)$$

$$ar(PQR) = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times QR \times PN \dots\dots\dots(ii)$$

Dividing (i) & (ii)

$$\frac{ar(ABC)}{ar(PQR)} = \frac{\frac{1}{2} \times BC \times AM}{\frac{1}{2} \times QR \times PN}$$

$$\Rightarrow \frac{ar(ABC)}{ar(PQR)} = \frac{BC \times AM}{QR \times PN} \dots\dots\dots(1)$$

In $\triangle ABM$ and $\triangle PQN$

$$\angle B = \angle C \quad [\because \text{Angles of similar triangles are equal}]$$

$$\angle M = \angle N \quad [\text{Being both } 90^\circ]$$

$$\triangle ABM \sim \triangle PQN \quad [\text{By AA similarity}]$$

$$\therefore \frac{AB}{PQ} = \frac{AM}{QN} \quad [\because \text{Corresponding sides of similar triangles are propotional}] \dots\dots\dots(2)$$

Now, from (1) and (2), we get

$$\begin{aligned} \frac{ar(ABC)}{ar(PQR)} &= \frac{BC \times AM}{QR \times PN} \\ \frac{ar(ABC)}{ar(PQR)} &= \frac{BC}{QR} \times \frac{AB}{PQ} \dots\dots\dots(3) \end{aligned}$$

Since, $\triangle ABC \sim \triangle PQR$

$$\therefore \frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{QR}$$

Putting these values in (3),

$$\frac{ar(ABC)}{ar(PQR)} = \frac{AB}{PQ} \times \frac{AB}{PQ} = \left(\frac{AB}{PQ} \right)^2$$

Again, using $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{QR}$, we get

$$\frac{ar(ABC)}{ar(PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{BC}{QR}\right)^2 = \left(\frac{AC}{PR}\right)^2$$

Hence proved.

47. Construct a triangle ABC in which AB = 4cm , BC = 6cm and $\angle B = 90^\circ$. Draw another similar triangle whose corresponding side is $\frac{3}{4}$ times the sides of $\triangle ABC$.

OR

Draw a tangent from a point P which is 10 cm away to a circle of radius 5cm.

Ans. Steps of construction:

- I. Construct $\triangle ABC$ in which AB = 4cm , BC = 6cm and $\angle B = 90^\circ$.
- II. At B draw an acute angle CBX below the base BC.
- III. Along BX, mark off points B_1, B_2, B_3, B_4 on BX such that $B_1B_2 = B_2B_3 = B_3B_4$.
- IV. Join B_4 with C and draw a line through B_3 parallel to line B_4C , to intersect the line BC at C' .
- V. Draw a line through C' parallel to the line AC to intersect line BA at A' .

Thus, $\triangle A'B'C'$ is the required triangle.

OR

Construct a pair of tangents from a point P which is 10 cm away to a circle of radius 5cm.

Ans. Steps of construction:

- 1) Draw a circle of radius 5 cm
- 2) Mark a point P, 10 cm away from the center B .
- 3) Draw a perpendicular bisector.
- 4) Wherever the perpendicular bisector intersect the line, take that point as center and draw a circle .
- 5) Join the points where the circle touches the circle with center B .

Hence, PQ and PR are the required tangents.

